

HEREDITARY DISORDERS OF GENOME INSTABILITY AND DNA REPAIR

AT A GLANCE

- Genome instability characterizes a large group of inherited disorders with prominent cutaneous abnormalities; many have an increased cancer risk.
- Genome instability is caused by impaired repair and/or maintenance of DNA.
- **Xeroderma pigmentosum (XP)** is a prototype disorder with defective repair of environmentally induced DNA damage and a greatly increased frequency of sunlight-induced cancers.
- Although these disorders are rare (on the order of 10^{-5} to 10^{-6}), heterozygote carriers of the affected genes may comprise several percent of the general population and may have an increased risk of neoplasia.

Overview

DNA is the carrier of genetic information, and its structure is constantly threatened by endogenous and exogenous damaging agents, including oxidative stress, ultraviolet (UV) radiation, ionizing radiation (e.g., x-rays), and chemical carcinogens. While cells possess multiple DNA repair pathways to maintain genomic integrity, failure in these systems can lead to abnormal cell function, cell death, or malignant transformation.

Persistent genome instability allows for the accumulation of mutations in critical genes within a single cell, a process that is often a prerequisite for carcinogenesis. This "mutator phenotype" significantly increases the likelihood of acquiring the multiple oncogenic mutations required for cancer development. The same genes that are mutated in hereditary genome instability syndromes

can also be impaired by acquired mutations in sporadic cancers, highlighting their central role in tumor suppression and DNA maintenance.

Clinical and Genetic Features

Hereditary disorders of genome instability are typically autosomal recessive and are characterized by defective DNA repair or DNA damage response mechanisms. These conditions frequently present with cutaneous manifestations, making dermatologists key players in their diagnosis and management. Most are also associated with an increased risk of malignancies, underscoring the critical role of genomic stability in preventing cancer.

Spontaneous vs. Induced Genome Instability

- **Spontaneous genome instability** is observed in: - **Bloom syndrome (BS)** - **Ataxia-telangiectasia (A-T)** - **Fanconi anemia (FA)**

These disorders show increased chromosome breakage and rearrangements even in unstimulated cells (e.g., peripheral blood or skin fibroblasts).

- **Induced genome instability** is a hallmark of:

- **Xeroderma pigmentosum (XP)**

In XP, DNA damage and genomic instability become apparent only after exposure to UV radiation or chemical carcinogens such as benzo[a]pyrene (found in cigarette smoke).

Clinical consequences—such as skin cancers and corneal damage leading to blindness—are largely preventable with rigorous avoidance of UV light, demonstrating the critical interaction between genetic susceptibility and environmental exposure.

Key Disorders and Mechanisms

Xeroderma Pigmentosum (XP)

- **Cutaneous features:** Extreme photosensitivity, poikiloderma (mottled pigmentation, atrophy, telangiectasia), and a high incidence of skin cancers (basal cell carcinoma, squamous cell carcinoma, melanoma).
- **Neurological features:** Sensorineural deafness, progressive neurodegeneration, and neuronal loss in some subtypes.
- **Cancer risk:** Drastically increased risk of UV-induced skin cancers; also elevated risk of ocular and central nervous system tumors.
- **Inheritance:** Autosomal recessive.
- **Molecular basis:** Mutations in genes involved in nucleotide excision repair (NER), including *XPA*, *XPB*, *XPC*, *XPD*, *XPE*, *XPF*, and *XPG*.
- **Impaired function:** Inability to repair UV-induced DNA lesions (e.g., cyclobutane pyrimidine dimers and 6-4 photoproducts).
- **Cellular phenotype:** Increased UV-induced cell death and mutagenesis.

Patients with XP who rigorously avoid UV radiation often do not develop skin cancer or blindness, illustrating the powerful impact of environmental modification in genetically susceptible individuals.

Other Major Disorders

While XP exemplifies defective repair of environmentally induced DNA damage, other syndromes involve broader defects in DNA damage response and replication:

- **Bloom syndrome:** Defective DNA helicase (*BLM* gene), leading to increased sister chromatid exchanges and genomic instability.
- **Ataxia-telangiectasia:** Mutations in *ATM*, a key regulator of the DNA damage response to double-strand breaks; associated with neurodegeneration, immunodeficiency, and cancer predisposition.

- **Fanconi anemia:** Defective interstrand crosslink repair via the FA/BRCA pathway; associated with bone marrow failure, congenital anomalies, and high risk of hematologic and solid tumors.

Public Health Implication

Although these disorders are individually rare, heterozygous carriers of the mutant alleles (e.g., parents of affected individuals) may represent a significant portion of the population. While typically asymptomatic, epidemiological data suggest that carriers may have a modestly increased risk of cancer, particularly in the context of environmental exposures.

This highlights the broader relevance of DNA repair pathways in general cancer susceptibility and supports the idea that subtle variations in DNA repair capacity across the population may influence individual cancer risk.

Conclusion

Hereditary disorders of genome instability provide critical insights into the mechanisms of DNA repair and their role in preventing cancer and maintaining tissue homeostasis. Their prominent cutaneous features make them visible markers of underlying genomic dysfunction. Understanding these syndromes not only aids in diagnosis and management but also informs broader cancer biology and prevention strategies, particularly the importance of gene-environment interactions.